

# Electric Potential Energy (completed)

## Goal.

- Understand where electric potential energy comes from: summing over pairs of point charges, or pushing a charge against a field.
- Use that to find the potential energy of an electric dipole in a uniform field.
- See that a system can have *negative* PE, and that “zero” PE is not always the lowest energy available.

## 1. Potential energy of two point charges (from the class prep)

Pulling a pair of point charges apart, or pushing them together, takes work — work that gets stored as potential energy.

$$U(q_1, q_2) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

- As  $r_{12} \rightarrow \infty$ ,  $U \rightarrow 0$ .
- For three or more charges, sum over every **pair** (not every charge).
- Opposite charges (+ and -):  $U$  is **negative** at any finite separation, and  $U \rightarrow 0$  only as  $r \rightarrow \infty$ . So  $U = 0$  is actually the **largest** value here — pulling the charges closer always finds a *lower* (more negative) PE.
- Like charges (same sign):  $U$  is **positive** at any finite separation, and again  $U \rightarrow 0$  as  $r \rightarrow \infty$ . Here  $U = 0$  is the **smallest** value — you can’t push like charges to a lower PE than infinitely far apart.

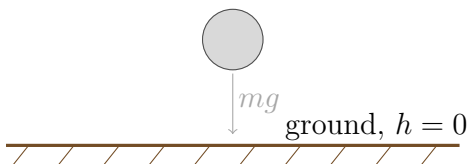
$$U_{\text{total}} = \frac{1}{4\pi\epsilon_0} \sum_{\text{pairs } (i,j)} \frac{q_i q_j}{r_{ij}}$$

**Check.** Two  $+q$  charges are brought from far apart to a separation  $r$ . Is  $U$  positive or negative? What about a  $+q$  and a  $-q$ ?

$+q$  and  $+q$ : positive (repulsive, energy stored pushing them together).  $+q$  and  $-q$ : negative (attractive).

## 2. Where is potential energy the lowest?

Draw a ball, and the ground below it.



**Check.** What is the ball's potential energy? Where is it the lowest?

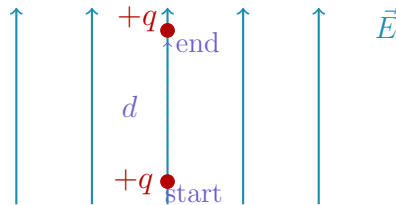
$U = mgh$ ; with  $h$  measured from the ground,  $U = 0$  there. As drawn, the ball can't go any lower than the ground — so here  $h = 0$  really is the lowest available PE (unlike the attractive point charges above, there's no “digging deeper” available).

This is a *uniform* gravitational field — not a hole to fall into. Raising the ball, moving it against the field lines, gives it *more* PE, and that PE is proportional to the distance you raise it.

$$\Delta U_{\text{grav}} = mgd$$

### 3. The same idea, for a charge in a uniform field

Now replace the ball with a positive charge  $q$ , and gravity with a uniform electric field  $\vec{E}$ . Draw the field, and the charge's starting and ending points, below.



The field pushes the charge with force  $qE$ . Push the charge a distance  $d$  against the field — the same motion as raising the ball.

$$\Delta U = qEd$$

- This is exactly like raising the ball: you do **positive** work pushing against the field, and that work becomes stored PE, proportional to **the distance  $d$** .
- Let the charge go, and it moves *with* the field instead: now  $\Delta U$  is **negative** — the field does the work, and spends the stored PE.

**Check.** A positive charge is released and moves along  $\vec{E}$ . Does its PE increase or decrease? Where did the energy go?

PE decreases — the charge is moving *with* the field, the way the ball falls rather than rises. The lost PE becomes kinetic energy.

## 4. The dipole in a field

A dipole is a  $+q$  and a  $-q$  separated by  $\vec{d}$  (dipole moment  $\vec{p} = q\vec{d}$ ), sitting in a uniform field  $\vec{E}$ , tilted at angle  $\theta$  between  $\vec{p}$  and  $\vec{E}$ . Each charge sits at a different place along  $\vec{E}$  — use §3's pushing idea on each charge, then add.

Split  $\vec{d}$  into a component along  $\vec{E}$  ( $d \cos \theta$ ) and a component across it. Only the along- $\vec{E}$  part matters (§3): the  $+q$  sits  $\frac{1}{2}d \cos \theta$  further along  $\vec{E}$  than center, the  $-q$  sits  $\frac{1}{2}d \cos \theta$  behind it. Adding the two charges' PE (each  $\propto -qEd$ , from §3, since moving *with*  $\vec{E}$  lowers PE) gives  $U = -qEd \cos \theta = -\vec{p} \cdot \vec{E}$ .

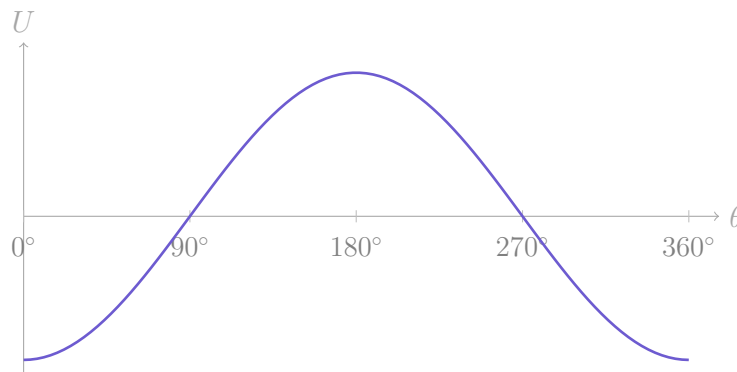
$$U(\theta) = -pE \cos \theta$$

- At  $\theta = 90^\circ$  (perpendicular to the field):  $U = 0$ .
- At  $\theta = 0^\circ$  (aligned with the field):  $U = -pE$  — which is **lower** than the  $\theta = 90^\circ$  value!
- At  $\theta = 180^\circ$  (anti-aligned):  $U = +pE$ .

**Check.** Which orientation is the most stable (lowest PE)? Is that the orientation where  $U = 0$ ? What does this have in common with the ball in the gravitational field?

$\theta = 0^\circ$  (aligned) is most stable, at  $U = -pE$  — *not* the  $U = 0$  orientation. Just like the ball:  $U = 0$  is just wherever we started measuring from, not necessarily the lowest energy the system can reach.

## 5. Sketch it: $U$ vs. $\theta$



Check your curve, and the point-charge and uniform-field results above, against the *Assembling Charges*, *PE vs. Separation*, and *Electric Dipole in a Field* sims — open their **Info** panels.